PassFrame tuition

Angles & 2D Shapes

Year 4 Pt 2 — 11+ Lesson Worksheet (PassFrame Tuition)

Worked examples — tutor talks through these first

Easy: What type of angle is 200°?
Method & answer: It is between 180° and 360°, so it is reflex.

Standard: Two angles on a straight line are 75° and x°. What is x?
Method & answer: Angles on a straight line add to 180°. x = 180 − 75 = 105°.

Advanced: A triangle has angles of 55° and 65°. What is the third angle?
Method & answer: Angles in a triangle add to 180°. 180 − 55 − 65 = 60°.

Easy / Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. What type of angle is 250°?
2. What type of angle is 88°?
3. Which of these is obtuse? A) 45° B) 90° C) 135° D) 15°
4. What do angles on a straight line always add up to?
5. What do angles around a point always add up to?
6. How many sides does a heptagon have?
7. Which shape has 9 sides? A) octagon B) nonagon C) decagon D) heptagon
8. How many sides does a decagon have?
9. Select ALL of these angles that are REFLEX (between 180° and 360°): A) 200° B) 90° C) 300° D) 170° E) 359°
10. The three angles inside any triangle always add up to how many degrees?
11. What type of triangle has no equal sides?
12. Which shape has 6 lines of symmetry? A) square B) regular hexagon C) equilateral triangle D) rectangle
13. What type of triangle has one angle greater than 90°?
14. What is the name for an angle greater than 180° but less than 360°?
15. How many sides does a triangle have?

Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. Two angles on a straight line are 84° and x°. What is x?
2. Three angles around a point are 120°, 100° and x°. What is x?
3. A triangle has angles of 45° and
6. In a triangle, two angles are equal at 48° each. What is the third angle?
7. Two straight lines cross, forming vertically opposite angles. If one angle is 63°, what is the angle vertically opposite it?
8. Select ALL of these that describe
11. A quadrilateral has angles of 90°, 60°, and two other equal angles. What is the size of each of the other two?
12. Which of these quadrilaterals has exactly one pair of parallel sides? A) trapezium B) rectangle C) rhombus D) square
13. The exterior angle of a triangle

75°. What is the third angle? A) 50° B) 60° C) 70° D) 80°

4. A quadrilateral has angles of 95°, 85° and 100°. What is the fourth angle?

5. Two equal angles lie around a point, and there are only two of them. What is the size of each?

a parallelogram: A) opposite sides are equal B) opposite sides are parallel C) all 4 sides are equal D) all 4 angles are 90° E) opposite angles are equal

9. An isosceles triangle has a base angle of 55°. What are the other two angles?

10. A right angle is split into two angles, one of which is 28°. What is the other angle?

equals the sum of the two opposite interior angles. If those are 35° and 50°, what is the exterior angle?

14. A triangle has one angle of 90° and another of 25°. What is the third angle?

15. The interior angles of a regular octagon sum to 1080°. What is each interior angle?

Standard / Hard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. Two angles on a straight line are (x+20)° and (2x+10)°. What is x, and what are the two angles?

2. Angles a and b lie on a straight line, and a = 2b. Find b.

3. A regular polygon has an exterior angle of 60°. How many sides does it have?

4. Select ALL of these that are TRUE about a kite: A) it has 2 pairs of adjacent equal sides B) all 4 sides are equal C) one pair of opposite angles are equal D) it always has 4 right angles E) it has at least 1 line of symmetry

5. A ladder leans against a wall, making an angle of 55° with the ground. What angle does it make with the wall?

6. A pizza is cut into equal slices, each with a 45° angle at the centre. How many slices are there in total?

7. Two parallel lines are cut by a transversal, forming co-interior angles of 70° and x°. Co-interior angles add to 180°. Find x.

8. A triangle has angles in the ratio 1:2:3. What is the size of each angle?

9. A quadrilateral's angles are all equal. What is the size of each angle, and which two shapes could it be?

10. A door is opened so it makes an angle of 55° with the wall. Through how many more degrees must it open to be at a right angle to the wall?

11. A regular polygon has an interior angle of 135°. How many sides does it have? (exterior angle = 180 − interior)

12. A triangular garden bed has one angle of 90° and another of 35°. What is the third angle, and what type of triangle is this?

13. Two straight roads cross at a junction, forming four angles. One angle measures 105°. What are the sizes of the other three angles?

14. A kite-shaped flag has two equal angles of 80° each and one other angle of 110°. What is the size of the fourth angle?

15. An isosceles triangle has an apex angle of 44°. What are the two base angles?